Regulating Maritime Autonomous Surface Ships in the Digital Era: International Maritime Law, Liability, and Insurance

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Abstract

The intersection of advanced technology, the legal framework, and implementation gives rise to a considerable challenge to the edifice of international maritime law in the context of maritime autonomous surface ships. The absence of crews and the complete automation of this new type of vessel raise questions about the applicability of current legal provisions. From a legal standpoint, the answer involves analyzing the adequacy of key international maritime conventions established under the International Maritime Organization concerning maritime autonomous surface ships. The analysis also examines the liability framework for maritime autonomous surface ships, as well as the effects that the introduction of these vessels may have on marine insurance. It is evident that amending the primary international maritime instruments will not yield a sufficient response. Traditional obligations cannot be fulfilled without crews on board. The adoption of a third-party liability regime, in this context, necessitates a model of strict liability. Moreover, the future regime of contractual liability must consider additional risks associated with damage to goods, the potential increase in liability limits, and the removal of certain exceptions that may allow carriers to evade liability. Marine insurance must adapt to new requirements, such as aligning the terms of standard insurance contracts with new risks like cyberattacks or the lack of adequate liability coverage.

Keywords**:** *maritime autonomous surface ships, international legal framework, the liability regime, maritime insurance.*

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1. Preliminary Remarks

The International Maritime Organization (IMO) has incorporated the integration of new, emerging, and advanced technologies into the objectives of its Strategic Plan. The changes resulting from technological advancements impact every facet of international maritime transport, encompassing ship design and the growing influx of data and information, as well as safety measures and the safeguarding of the marine environment[2]. The shipping industry is increasingly susceptible to cyberattacks, primarily because of its heightened dependence on digital technologies[3]. In this context, the effective regulation of international maritime transport relies on two key factors: firstly, the adaptation of the existing legislative framework to accommodate new technologies; and secondly, the adoption of new legislative instruments alongside the facilitation of technology transfer[4]. In 2021, the International Maritime Organization (IMO) conducted an analysis of the regulatory framework to examine pertinent treaties related to ship safety, with the objective of evaluating how maritime autonomous surface ships might be regulated[5]. Maritime autonomous surface ships are ships with varying degrees of automation, ranging from partially automated systems that assist the human crew to fully autonomous systems capable of handling all aspects of a ship's operation without the need for human intervention[6]. At the legislative level, the urgent priorities stemming from the operation of maritime autonomous surface ship-class vessels involve amending the pertinent IMO instruments, establishing suitable terminology, and prioritising the resolution of legislative gaps[7]. In 2022, the International Maritime Organization (IMO) adopted a roadmap that outlines the development of a goal-based instrument, which will take the form of a non-binding code for autonomous surface vessels[8]. At the same time, efforts at the European Union (EU) level have resulted in the adoption of the EU Operational Guidelines on trials of autonomous surface ships. These

[2] International Maritime Organization, Resolution A.1196(34) Adopted on 3 December 2025 (Agenda item 8(a)), Revised Strategic Plan for the Organization for the six-year period 2024 to 2029. https://www.imo.org/en/about/strategy/pages/default.aspx, accessed on 13 March 2026.
[3] *Ibid.*
[4] *Ibid*
[5] International Maritime Organization: Autonomous ships: Regulatory scopind exercise completed https://www.imo.org/en/mediacentre/pressbriefings/pages/massrse2021.aspx, accessed on 13 March 2026.
[6] IMO MSC.1/Circ.1638, 3 June 2021 Outcome of the Regulatory Scoping Exercise for the use of Maritime Autonomous Surface Ships (MASS), https://www.imo.org/en/mediacentre/pressbriefings/pages/massrse2021.aspx, accessed on 13 March 2026.
[7] *Ibid.*
[8] International Maritime Organization, Maritime Safety Committee (MSC 105), 20-29 April 2022, https://www.imo.org/en/mediacentre/meetingsummaries/pages/msc-105th-session.aspx, accessed on 13 March 2026.

guidelines are designed to assist the national administrations of Member States in ensuring that the testing of autonomous surface vessel technologies is carried out safely and in a manner that safeguards the environment[9].

This study aims to analyse the adequacy of legislation concerning autonomous surface vessels by exploring several key areas: international conventions related to maritime autonomous surface ships; both contractual and non-contractual aspects associated with the utilisation of these vessels; and the implications of international marine insurance for maritime autonomous surface ships. The methods employed are quantitative, focusing particularly on the logical method and the prospective method.

2. International Instruments That Address the Legal Aspects of Maritime Autonomous Surface Ships

Maritime autonomous surface ships are classified into four categories based on their level of autonomy. Degree one: Ship with automated processes and decision support- the crew must be present to operate and monitor the ship's systems and functions. Degree two: Remotely controlled ship with seafarers on board: The ship is controlled and operated from another location. Degree three: Remotely controlled ship without seafarers on board: Degree four: Fully autonomous ship: The operating system of the ship is able to make decisions and determine actions by itself[10]. Applying the provisions of international maritime conventions to autonomous surface ships poses a legislative challenge for autonomy degrees three or four. From a legal standpoint, it is important to highlight that maritime autonomous surface ships currently lack legal definitions under international maritime conventions. The relevant international maritime conventions, therefore, do not apply[11]. A second important issue is the absence of crew members on maritime autonomous surface ships classified at degrees three and four. In the case of the United Nations Convention on the Law of the Sea (UNCLOS), it is unclear how the requirement that ships "that each ship is in the charge of a master and officers who possess appropriate qualifications, in particular in seamanship, navigation, communications and marine engineering, and that the crew is appropriate in qualification and numbers for the type, size, machinery and

[9] EU Operational Guidelines for safe, secure and sustainable trials of Maritime Autonomous Surface Ships (Mass), https://transport.ec.europa.eu/system/files/2020-11/guidelines_for_safe_mass.pdf, accessed on 13 March 2026.

[10] International Maritime Organization. MSC 100/20/Add.1 Annex 2: Framework for the Regulatory Scoping Excercise for the Use of Maritime Autonomous Surface Ships (MASS) https://maiif.org/wp-content/uploads/2019/06/MSC-100_20-Annex-20-1.pdf, accessed on 13 March 2026.

[11] Paul Dean and Henry Clack, „Autonomous Shipping and Maritime Law". In Baris Soyer, Andrew Tettenborn (eds) *New Technologies Artificial Inteligence and Shipping Law in the 21st Century*, Informa Law from Routledge, 2021, p. 70.

equipment of the ship"[12] The obligation "to assist persons in distress at sea" cannot be fulfilled without the presence of the captain and the crew[13]. According to the UNCLOS, flag states possess exclusive jurisdiction over their vessels while they are on the high seas[14]. In the absence of a crew on board, the exercise of the flag state's rights and obligations, which are founded on the genuine link between the state and the flag, may be compromised[15]. In cases where remote operators or algorithms control vessels from a command centre located in a state apart from the flag state, enforcement measures taken by flag states in the event of non-compliance may conflict with the territorial jurisdiction of that state over the command centre and the operators located therein[16].

The International Convention for the Safety of Life at Sea (SOLAS) of November 1, 1974, defines ships by reference to the category to which they belong, namely passenger ships, cargo ships, tankers, and fishing vessels, without reference to any other categories of ships[17]. The application of the provisions of the SOLAS Convention to maritime autonomous surface ships necessitates amending the Convention's provisions to extend the definitions to cover such vessels. The absence of a crew on board necessitates amendments to various provisions, including those concerning structure, subdivision, stability, machinery, and electrical installations (Chapter II-1); functional requirements for fire detection and control systems that rely on the operation by the ship's crew (Chapter II-2); processes for evacuating passengers from a ship and rescuing individuals from the water (Chapter III); technical requirements for remote control stations (Chapter IV); the roles, responsibilities, and qualifications of crews; the transportation of equipment and associated performance standards; the transfer of information and communication between the ship, shore, and remote control stations (Chapter V); as well as the implementation of emergency procedures for spill or fire conditions involving onboard cargo and procedures for maintaining safety under normal conditions (Chapter VI)[18]. The SOLAS Convention does not contain provisions regarding remote control stations, remote operators, connectivity, and cyberattacks within the framework of ship safety management (Chapter IX)

[12] Article 94 para. 4 lit. b United Nations Convention on the Law of the Sea www.un.org/depts/los/convention_agreements/texts/unclos/unclos_e.pdf, accessed on 13 March 2026.
[13] *Ibid.* Article .98. para.1 letter a).
[14] *Ibid.* Article 92.
[15] Young-Hun Min and Jung-Hwan Choi, „Unmanned Ships and Flag State Jurisdiction". In Chong-Ju Chae, Raphael Baumler (eds.) *Maritime Autonomous Surface Ships (MASS) - Regulation, Technology, and Policy. Three Dimensions of Effective Implementation,* WMU Studies in Maritime Affairs, volume 11, Springer, 2024, p. 31.
[16] *Ibid*., p. 37.
[17] Regulation 2 International Convention for the Safety of Life at Sea (SOLAS) https://www.imorules.com/SOLAS.html, accessed on 13 March 2026.
[18] International Maritime Organization, MSC.1/Circ.1638, 3 June 2021 Outcome of the Regulatory Scoping Exercise for the use of Maritime Autonomous Surface Ships (MASS), https://www.imo.org/en/mediacentre/pressbriefings/pages/massrse2021.aspx, accessed on 13 March 2026.

or the regulation of remote-control operational centers where on-board control or manual operation is considered a mandatory requirement (Chapter XI-2)[19].

The International Convention on Standards of Training, Certification, and Watchkeeping for Seafarers (STCW) seeks to establish global standards for the certification, training, and professional development of seafarers, as well as for their performance of watchkeeping duties[20]. In the absence of a crew on board, the provisions related to certificates and endorsements[21], training and assessment[22], company responsibilities[23] and the execution of watchkeeping duties[24] are not applicable to maritime autonomous surface ships classified as degrees three and four. The instrument regulating the legal framework for maritime autonomous surface ships must consider training, certification, and competency requirements in accordance with STCW standards[25].

The Convention on the International Regulations for Preventing Collisions at Sea, 1972 (COLREG), regulates technical aspects concerning steering and navigation, lights and markings, sound and light signals, exemptions, and the verification of compliance with the provisions of the Convention[26]. The COLREG Convention's scope permits the inclusion of maritime autonomous surface ships[27]. Legal difficulties arise from the provisions concerning the responsibility of the master or crew thereof, from the consequences of any neglect to comply with these rules, or from the neglect of any precautions which may be required by the ordinary practice of seamen or by the special circumstances of the case[28]. From a technical perspective, maritime autonomous surface ships need to be modified to comply with the convention's requirements concerning collision risk[29] and avoidance measures[30].

Other international maritime conventions, including the Maritime Labour Convention 2006, become irrelevant without a crew on board. International maritime conventions related to marine pollution necessitate a distinct form of

[19] *Ibid.*
[20] International Maritime Organization STCW/CONF.2/33, 1 July 2010 Resolution 1 The Manila Amendments to the annex to the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers (STCW), 1978.
[21] *Ibid* Regulation I/2.
[22] *Ibid* Regulation I/6.
[23] *Ibid* Regulation I/14.
[24] *Ibid* Regulation VIII/1- Regulation VIII/2.
[25] Paul Dean and Henry Clack, *op.cit.*, p. 78.
[26] Convention on the International Regulations for Preventing Collisions at Sea, 1972 https://treaties.un.org/doc/publication/unts/volume%201050/volume-1050-i-15824-english.pdf, accessed on 13 March 2026.
[27] *Ibid*. Regulation 1.
[28] *Ibid*. Regulation 2.
[29] *Ibid.* Regulation 7.
[30] *Ibid.* Regulation 8.

clarification. The International Convention on Civil Liability for Bunker Oil Pollution Damage (BUNKER) defines "vessel" as any seagoing ship and any type of seagoing craft.[31] This Convention applies to maritime autonomous surface ships, except those that do not carry bunkers and utilize an alternative energy source for propulsion[32]. According to the convention, a shipowner is defined to include not only the registered owner but also the bareboat charterer, the manager, and the operator of the ship. It is challenging to ascertain whether a shore-based operator engaged in the remote management of a vessel, or a developer of operating systems can restrict their liability according to the existing language of the convention[33]. Similar to the International Convention on Civil Liability for Bunker Oil Pollution Damage, the International Convention on Civil Liability for Oil Pollution Damage (CLC, 1992) outlines the conditions under which a shipowner may be exempt from liability: (a) the damage: resulted from an act of war, hostilities, civil war, insurrection or a natural phenomenon of an exceptional, inevitable and irresistible character; or (b) was wholly caused by an act or omission done with intent to cause damage by a third party, or (c) was wholly caused by the negligence or other wrongful act of any Government or other authority responsible for the maintenance of lights or other navigational aids in the exercise of that function[34]. The application of the previously mentioned conventions to maritime autonomous surface ships requires the involvement of classification societies for non-compliant standards or IT specialists to address technical malfunctions within the given exceptions[35]. The significance of these new actors should be examined in relation to other conventions addressing marine pollution, including the International Convention on Liability and Compensation for Damage in Connection with the Carriage of Hazardous and Noxious Substances by Sea, 2010 (2010 HNS Convention)[36]. The International Convention for the Prevention of Pollution from Ships 1973/78 (MARPOL) affects the format of financial guarantee certificates required for maritime autonomous surface ships, as well as the procedures for their inspection by port state control[37].

[31]Article 1 International Convention on Civil Liability for Bunker Oil Pollution Damage (BUNKER), https://www.jus.uio.no/english/services/library/treaties/06/6-07/bunkers-convention.html, accessed on 13 March 2026.
[32] Paul Dean and Henry Clack, *op. cit.*, p.82,
[33] Wangwang Xing, „Contemplating Maritime Autonomous Surface Ships (MASS) under the international law on ship-source pollution", *Marine Pollution Bulletin* 207 (2024) 116884, DOI: 10.1016/j.marpolbul.2024.116884.
[34] Article 3 International Convention on Civil Liability for Oil Pollution Damage, 1992, https://iopcfunds.org/wp-content/uploads/2018/06/Text-of-Conventions_e.pdf, accessed on 13 March 2026.
[35] Wangwang Xing, *op. cit.*, p. 10.
[36] *Ibid.*
[37] *Ibid.*

3. The Liability Regime for Maritime Autonomous Surface Ships

Traditionally, the carrier's contractual liability is governed by the Hague-Visby Rules, the Hamburg Rules, or the Rotterdam Rules. If these instruments apply to maritime autonomous surface ships, it is essential to address several important issues: who is responsible for the vessel's seaworthiness, what defenses can the carrier invoke to be exempt from liability, and who may benefit from the limitation of liability. The Hague-Visby Rules stipulate that: The carrier shall be bound before and at the beginning of the voyage to exercise due diligence to: (a) make the ship seaworthy; (b) properly man, equip and supply the ship; (c) make the holds, refrigerating and cool chambers, and all other parts of the ship in which goods are carried, fit and safe for their reception, carriage and preservation[38]. The carrier is responsible for any failure to meet its obligations when, in the case of maritime autonomous surface ships, those obligations are delegated to experts or subcontractors[39]. The carrier does not possess the requisite expertise to assess whether the systems and programmes developed by software developers guarantee the proper functioning of maritime autonomous surface ships[40]. The adoption of new technologies for the carriage of goods on maritime autonomous surface ships does not exempt operators from adhering to the care requirements outlined in the Hague-Visby Rules[41].

Article IV (2) of the Hague - Visby Rules outlines the defences available to a carrier seeking to evade liability. (a) Act, neglect, or default of the master, mariner, pilot, or the servants of the carrier in the navigation or in the management of the ship. (b) Fire, unless caused by the actual fault or privity of the carrier. (c) Perils, dangers and accidents of the sea or other navigable waters. (d) Act of God. (e) Act of war. (f) Act of public enemies. (g) Arrest or restraint of princes, rulers or people, or seizure under legal process. (h) Quarantine restrictions. (i) Act or omission of the shipper or owner of the goods, his agent or representative. (j) Strikes or lockouts or stoppage or restraint of labour from whatever cause, whether partial or general. (k) Riots and civil commotions. (l) Saving or attempting to save life or property at sea. (m) Wastage in bulk of weight or any other

[38] Article III r 1 International Convention for the Unification of Certain Rules of Law relating to Bills of Lading ("Hague Rules") as Amended by the Brussels Protocol 1968 ("Hague-Visby Rules") and by the Brussels Protocol 1979, https://www.fog.it/convenzioni/inglese/visby-1979.htm, accessed on 13 March 2026.

[39] Richard Williams, „Unmanned ships – a challenge to the current international regime regulating the carriage of goods?". In Baris Soyer, Andrew Tettenborn (eds), *Ship Operations New Risks, Liabilities and Technologies in the Maritime Sector*, Informa Law from Routledge, 2022, p. 225.

[40] *Ibid.*

[41] Frank Stevens, „Carrier Liability for Unmanned Ships. Goodbye Crew, Hello Liability" in Baris Soyer, Andrew Tettenborn (eds) *New Technologies Artificial Inteligence and Shipping Law in the 21st Century*, Informa Law from Routledge, 2021, p.160.

loss or damage arising from inherent defect, quality or vice of the goods. (n) Insufficiency of packing. (o) Insufficiency or inadequacy of marks. (p) Latent defects not discoverable by due diligence. (q) Any other cause arising without the actual fault or privity of the carrier, or without the fault or neglect of the agents or servants of the carrier. In this case, it is necessary to ascertain whether a person operating a maritime autonomous surface ship from shore can be regarded as an agent or representative of the carrier[42]. If legal proceedings are initiated against the company engaged by the carrier to supply the hardware systems for operating maritime autonomous surface ships, those parties are regarded as independent contractors and cannot invoke exemption from liability under the Hague-Visby Rules[43]. A similar principle applies to the limitation of liability in legal proceedings concerning the loss of or damage to goods. The Hague - Visby Rules provide for the carrier's right to limit its liability[44]. The benefit of the limitation of liability is forfeited if it can be demonstrated that the damage arose from an act or omission of the carrier, undertaken with the intent to cause the damage, or through negligence while being aware that such damage could occur[45]. It seems improbable that, within the framework of maritime autonomous surface ships, an independent contractor on land (a company) entrusted with tasks by the carrier would qualify for the protection afforded by the Hague-Visby Rules[46]. An independent contractor can only extend the benefit of the limitation of liability by including a Himalaya clause in the bill of lading[47].The use of maritime autonomous surface ships will give rise to additional risks regarding damage to cargo, an increase in liability limits, or the elimination of certain exceptions that may be invoked to exempt the carrier from liability under the Hague-Visby Rules[48]. In addition, it will be essential to redistribute responsibilities by engaging other stakeholders, including rescuers[49]. Paradoxically, achieving full operational autonomy for maritime autonomous surface ships will require addressing the traditional require-

[42] Richard Williams, *op. cit.* p. 229.
[43] *Ibid.* p.230.
[44] Art. IV r 5 (a) 1 International Convention for the Unification of Certain Rules of Law relating to Bills of Lading ("Hague Rules") as Amended by the Brussels Protocol 1968 ("Hague-Visby Rules") and by the Brussels Protocol 1979, https://www.fog.it/convenzioni/inglese/visby-1979.htm.
[45] *Ibid.* art. IV r 5 (e).
[46] Richard Williams, *op. cit*. p.235.
[47] Simon Baughen, „Who is the Master Now? Regulatory and Contractual Challenges of Unmanned Vessels". In Baris Soyer, Andrew Tettenborn (eds.) *New Technologies Artificial Inteligence and Shipping Law in the 21st Century*, Informa Law from Routledge, 2021, p.140.
[48] Mikis Tsimplis, „On the Carriage of Goods on Autonomous, Remotely Controlled, Crewed and Uncrewed Ships". In Stephen Girvin, Vibe Ulfbec (eds.), *Maritime Organisation, Management and Liability A Legal Analysis of New Challenges in the Maritime Industry*, Hart Publishing Bloomsbury Publishing, 2021, p. 305.
[49] *Ibid*.

ments associated with the transport of goods by sea from a technological perspective[50]. Contracts for the chartering of ships for the transport of goods require reassessment. The shipping industry must adapt the terms of time charter (NYPE) and voyage charter (Gencon) contracts to reflect the reality of transporting goods using maritime autonomous surface ships[51]. The adoption of standardized contracts for maritime autonomous surface ships is essential to prepare for the eventual widespread use of this type of vessel in international shipping[52].

The introduction of maritime autonomous surface ships necessitates the establishment of a strict liability regime for tortious liability, aligning with the majority of international maritime conventions regarding marine pollution[53]. Conventions like the International Convention on Civil Liability for Oil Pollution Damage (1992) and the Convention on Limitation of Liability for Maritime Claims (1976)[54] impose strict liability on shipowners. The doctrine proposes establishing a legal framework for maritime autonomous surface ships that mirrors existing regulations. Liability should be assigned to the registered owner of the vessel[55]. He has the right to pursue a legal action for tort liability due to negligence against either the software manufacturer or the developer[56]. The party accountable for maritime autonomous surface ships is strictly liable for any losses resulting from cyberattacks on the vessels[57].

4. Insurance and Maritime Autonomous Surface Ships

Marine insurance closely ties into the international shipping industry. The emergence of maritime autonomous surface ships necessitates a reevaluation of marine insurance contracts, as these ships introduce new categories of risk[58].

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[50] *Ibid*.

[51] Grant Hunter, „Standard contracts for the MASS (es) – charter parties and other agreements for autonomous ships". In Baris Soyer, Andrew Tettenborn (eds), *Ship Operations New Risks, Liabilities and Technologies in the Maritime Sector*, Informa Law from Routledge, 2022, p. 219.

[52] *Ibid*.

[53] International Convention on Civil Liability for Oil Pollution Damage, 1992, https://iopcfunds.org/wp-content/uploads/2018/06/Text-of-Conventions_e.pdf, accessed on 13 March 2026.

[54] Convention on limitation of liability for maritime claims, 1976 (with final act). Concluded at London on 19 November 1976, https://treaties.un.org/doc/Publication/UNTS/Volume%201456/volume-1456-I-24635-English.pdf, accessed on 13 March 2026.

[55] Barış Soyer, „Autonomous Vessels and Third-Party Liabilities - The Elephant in the Room". In Baris Soyer, Andrew Tettenborn (eds) *New Technologies Artificial Inteligence and Shipping Law in the 21st Century*, Informa Law from Routledge, 2021, p.112.

[56] *Ibid*. p. 113.

[57] *Ibid*. p. 111.

[58] Manuel Varela Chouciño, Jonatan Echebarria Fernández, and Johanna Hjalmarsson, „Insurance of Marine Autonomous Surface Ships: Risk Allocation, Seaworthiness and Technological Challenges from an Underwriter's Perspective". In Tafsir Matin Johansson, Jonatan Echebarria Fernández, Dimitrios Dalaklis, Aspasia Pastra, Jon A. Skinner (eds.), *Autonomous Vessels in Maritime Affairs, Law and Governance Implications,* Palgrave Macmilan, 2023, p. 316.

The use of maritime autonomous surface ships may impact key concepts, including the accurate presentation of risk in insurance policies, the seaworthiness of ships, the manning requirements for those ships, and the processes involved in watchkeeping[59]. Implied warranties, including seaworthiness, necessitate a reassessment to ascertain if traditional inspection methods are applicable to maritime autonomous surface ships[60]. It remains uncertain whether the standard insurance terms for constructing ships and conducting sea trials, along with Hull and Machinery (H&M) insurance and cargo insurance for maritime autonomous surface vessels, will offer sufficient coverage[61]. To what extent do emerging security risks, such as piracy and cyberattacks, impact traditional marine insurance contracts? For instance, a cyberattack may lead to a collision, cause property damage, result in pollution, or cause a ship to run aground[62]. Generally, cyberattacks are included under P&I Club insurance coverage, except when they are classified as acts of terrorism or acts of war[63]. The exposure of shipowners and managers to third-party liability is likely to increase. One thing is clear: as reliance on technology grows, it will become increasingly challenging to ascertain the shipowner's fault in the event of a collision[64]. Liability for incidents involving maritime autonomous surface ships will affect both prices and the overall risk structure[65].

5. Conclusions

The international instruments adopted under the auspices of the IMO lack definitions that would enable the application of those conventions to maritime autonomous surface ships. The introduction of maritime autonomous surface ships necessitates a reconfiguration of the existing legal framework to accommodate the new technical and legal realities introduced by their presence. The presence of small crews on maritime autonomous surface ships does not create significant issues; however, achieving full autonomy and operating without a crew introduces substantial legislative challenges at the international level. The legislative impact encompasses all legal aspects related to the operation of autonomous surface vessels, including the jurisdiction of contracting states, navigational safety, and both contractual and tort liability in instances of pollution or

[59] *Ibid*, p. 316, 318.
[60] *Ibid.*, p. 322.
[61] *Ibid.*, p. 324.
[62] UK P&I Club Cyber risks and P&I Insurance, https://safety4sea.com/wp-content/uploads/2018/03/UK-PI-Club-Cyber-Risks-and-PI-insurance-2018_03.pdf, accessed on 13 March 2026.
[63] *Ibid.*
[64] Luci Carey, „Autonomous ships and Hull and Machinery Insurance" in Stephen Girvin, Vibe Ulfbec (eds.), *Maritime Organisation, Management and Liability a Legal Analysis of New Challenges in the Maritime Industry*, Hart Publishing Bloomsbury Publishing, 2021, p. 258.
[65] Paul Dean and Henry Clack, *op. cit.*, p. 85.

collision, as well as marine insurance. From a legislative standpoint, one potential solution is to amend all relevant international maritime conventions, though this process is notably challenging to execute. An alternative solution would be to create a single instrument dedicated exclusively to the legal issues involved in the operation of maritime autonomous surface ships. Certain aspects may be unprecedented, particularly regarding the legal liability of shore-based personnel who remotely operate these vessels.

The legal regime governing maritime autonomous surface ships must be brought into line with relevant international maritime conventions regarding liability to third parties. Insurance contracts and maritime insurance in general must take into account the regulation of new risks, such as cyberattacks, and adopt the necessary legislative solutions. While many issues await resolution through judicial practice, the delay in adopting legislative measures leaves room for ambiguity. Another argument in favor of adopting international instruments as quickly as possible is the need for legislative uniformity at the international level as the use of this new type of vessel becomes more widespread.

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